COMP534 – Assessment 2
Deadline: 17th of April 2026

As deep neural networks are exploited in different ways and in various different domains, how to properly create and train a neural network to achieve the best outcome for a specific task has become very important. In this assignment, your aim is to build and train neural network models to solve a classification problem.

Specifically, you will work with a lung X-ray dataset, **available as a ZIP file on Canvas**, to build a model that identifies different types of lung conditions. The dataset contains approximately 3,500 images with a resolution of 299x299 pixels and is unevenly distributed across 3 classes:

1. Normal: X-ray images of healthy lungs
2. Opacity: X-rays showing varying degrees of lung abnormalities
3. Pneumonia: images associated with viral pneumonia cases

You **will need to submit your code and a report**. For the code, please **use the provided Jupyter notebook template**, which is also available for download on Canvas. The sections of this notebook match the sections your report should have, which are:

1. Data Management
    a. In the Jupyter Notebook:
        i. define an appropriate experimental protocol (such as k-fold, cross validation, etc);
        ii. create the dataloader to load the data; remember to include here any normalization, data augmentation, or other technique used to pre-process the data;
    b. In the report:
        i. detail the experimental protocol, i.e., how you split the data;
        ii. explain any pre-processing (normalization, etc) performed in the input data;
        iii. detail any data augmentation.
2. Neural Networks
    a. In the Jupyter Notebook:
        i. propose **your own** Convolutional Neural Network (CNN) to tackle the problem, aiming to make it as efficient (in terms of trainable parameters) as possible;
        ii. define at least one **existing** CNN (such as AlexNet, VGG, ResNet, DenseNet, etc) to tackle the problem;
        iii. define the necessary components to train the networks (that is, loss functions, optimizers, etc);
        iv. train your proposed architecture **from scratch** using your training set;
        v. train the **existing** architecture using **at least 2 different strategies** (i.e., trained from scratch, fine-tuning, feature extractor, etc);

vi. for all training procedures, separately plot the loss and accuracy with respect to the epoch/iteration.

b. In the report:

i. briefly describe the concept behind your proposed network;
ii. describe the selected existing architecture and training strategies;
iii. explain your decisions regarding the training components (loss functions, optimizers, etc);
iv. report table of the results (using appropriate metrics). Compare the methods and exploited training strategies and define the best model.

3. Evaluate models

a. In the Jupyter Notebook:

i. evaluate the model (the best one you obtained in the above stage) on the appropriate set.

b. In the report:

i. report results for the appropriate set using **at least** 3 different metrics **suitable** for your problem. The selected metrics should capture different aspects of model performance (avoid reporting closely related measures);
ii. plot and discuss the confusion matrix.

Submission and Team Size

For this assignment, you can work independently, in pairs, or in a group of **THREE** or **FOUR**. You are free to choose your own team members, and the size of the team will have no impact on the final mark given. Also, all members of a team will get the same grade.

The deadline is the 17th of April at 5PM

You will need to submit your completed assessment **on Canvas**. Submit **a single zip file** which includes a report (in pdf) and the python code (in .ipynb). Your report should **not be more than 1200 words long**.

Your submission (that is, your .zip file) should have the name in the format of: "studentIDOfMember1_ studentIDOfMember2_studentIDOfMember3.zip" (example for a group of 2 students: 201712345_201812345.zip)
Please, include the student IDs of all members. Also, all members need to make the final submission on Canvas.

Rubric

The project assessment against the following criteria:

1. Python code (**60%**)

a. Correct definition of the data split / protocol (**5%**)

   b. Appropriate dataloader with correct normalization, data augmentation, etc (**13%**)
   c. Correct definition of network architectures (**20%**)
   d. Correctly define the necessary training components, train the models, and plot graphs (**17%**)
   e. Test using the appropriate set (**5%**)
2. Report (**40%**)
   a. Clearly and briefly describe all details of the experimental protocol (**5%**)
   b. Explain the details of your dataloader, including any pre-processing performed (such as normalization, data augmentation, etc) (**12%**)
   c. Briefly describe the (concept of the) exploited network architectures and strategies, as well as the components required for training (**13%**)
   d. Report and discuss the training results table using appropriate metrics - clearly define the best model and justify/discuss (**5%**)
   e. Present and discuss a table and a confusion matrix with the final results using the best model in the appropriate set and with appropriate metrics (**5%**)